



UKUDLA Data Science Certificate

Data Visualization

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UKUDLA
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for Sustainable and Resilient Food
Systems and Applied Agricultural
and Food Data Science



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Prepared data do not reveal their patterns by themselves

A table preserves exact values

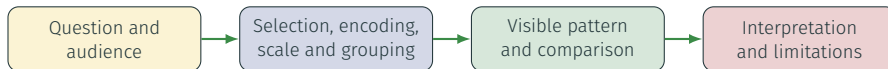
- study units and occasions are explicit;
- units and variables are documented;
- observations can be queried; and
- computations are reproducible.

A visual representation can reveal

- distributions and unusual values;
- differences between groups or units;
- change over time; and
- relationships between variables.

Data visualization maps documented observations to visual properties so that patterns can be inspected and communicated.

A chart is an analytical argument



- Exploratory graphics support investigation and iteration.
- Explanatory graphics focus attention on a defensible message.
- Both depend on choices about inclusion, aggregation and visual encoding.
- Neither turns an association into a causal effect.

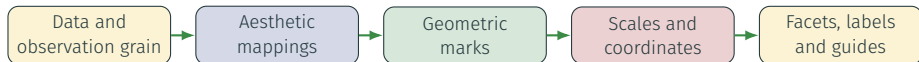
This session moves from **motivation**, through essential **concepts**, to a reproducible **application** in an analytical workflow.

Start with the analytical question

Question	Relevant structure	Possible graphic
How are values distributed?	One quantitative variable	Histogram, density plot, boxplot
How do groups compare?	Quantitative value and category	Aligned points, boxplots, faceting
How does a measure change?	Quantitative value ordered by time	Line or point plot
How do two measures vary together?	Two quantitative variables	Scatterplot
Where does a measure occur?	Measure and valid spatial geometry	Map

Choose a graphic from the question and variable types—not from a menu of decorative chart forms.

Build a graphic from explicit components



- Map variables to position, colour, shape, size or grouping.
- Select marks—points, lines, bars or areas—that support comparison.
- Use faceting when small multiples clarify group differences.
- Add labels and annotations that make definitions and units visible.

In `ggplot2`, every visible element should follow from an explicit layer or mapping.

Use the strongest encodings for important comparisons

Usually easier to compare

1. position on a common scale;
2. position on aligned scales;
3. length; and
4. angle or area.

Use with care

- colour hue distinguishes categories;
- colour lightness can show order;
- size is read through area, not radius;
- shape supports few categories; and
- redundant cues improve accessibility.

Encode the quantity that matters most with position when possible; reserve colour and size for distinctions they can carry reliably.

Scales and aggregation change what the viewer sees

Choice	Risk	Review question
Axis limits	Differences appear exaggerated or observations disappear	Does the scale support an honest comparison?
Transformation	Distances acquire a different interpretation	Is a log or other scale named and justified?
Aggregation	Within-group variation becomes invisible	What grain and summary rule produced each mark?
Binning	Distribution shape changes with break placement	Are conclusions stable across reasonable bins?
Overplotting	Dense observations conceal frequency	Would transparency, jitter, bins or facets help?
Free facet scales	Shapes are comparable but magnitudes are not	Does each panel need the same quantitative scale?

Never hide a consequential filtering, transformation or aggregation decision in plotting code.

Make meaning visible

- informative title and subtitle;
- axis labels with units;
- readable legends or direct labels;
- source and aggregation note; and
- annotation of the intended comparison.

Make the figure usable

- sufficient contrast and text size;
- colour-blind-safe palettes;
- meaning not carried by colour alone;
- appropriate dimensions and file type; and
- reproducible code and stable output path.

A self-contained figure states what is shown, how it was measured and what it cannot establish.

Ask complementary questions of prepared data

One outcome across a study

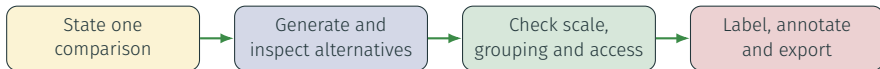
- distribution of the outcome;
- differences between treatments, sites or groups;
- change over time or measurement order; and
- unusual units or measurement occasions.

Outcome and a second variable

- distribution and patterns of the second variable;
- outcome versus predictor;
- differences across relevant groups; and
- possible non-linearity or overplotting.

Each chart should state its observation grain: one mark may represent a sample, plot, visit, reported entity, group summary or aggregated bin.

Turn an exploratory plot into a communication graphic



- Begin with a trend, distribution or relationship plot tied to one question.
- Compare faceting, grouping and scale choices against the intended question.
- Remove decoration that does not carry information.
- Add units, study context, alignment rules and a non-causal interpretation.
- Save the figure through code rather than manual export.

Key message: every visual choice shapes interpretation

Reproducible artifacts

- plotting script;
- exploratory figure set;
- one communication-ready figure;
- documented output paths; and
- source data and package environment.

Review evidence

- question matches the graphic;
- grain, units and scales are visible;
- grouping and aggregation are justified;
- design remains accessible; and
- interpretation respects limitations.

Visualization reveals and communicates patterns; the next session quantifies distributions and associations with descriptive statistics.

Show the evidence clearly—and make every consequential visual choice inspectable.

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